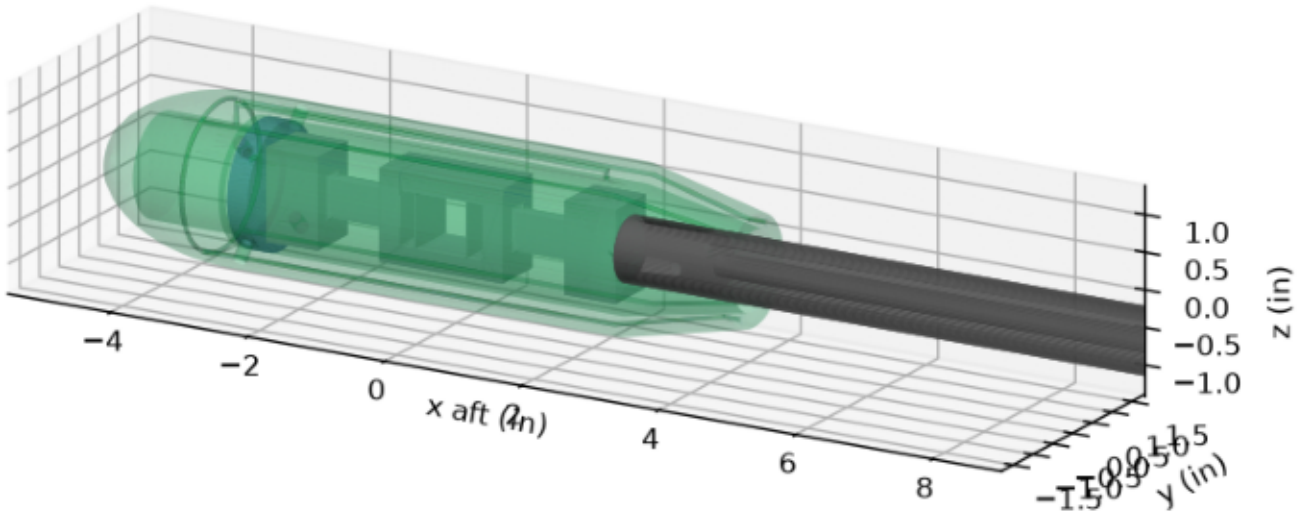


FLIGHT ASSEMBLY (CAD RENDER, NTS)

Flight assembly: instrumented 2.25 in pod (transparent), puck, 6-component balance, sting



NOTES:

1. GAUGE AND WIRE THE BALANCE (24 GAUGES, 6 BRIDGES) BEFORE ASSEMBLY; SEE THE GAUGE LAYOUT DRAWING.
2. PUCK TO BALANCE FRONT FACE: 2 DOWELS + 2 SCREWS. STING ON THE SPIGOT: CROSS PIN + 2 SET SCREWS.
3. WIRES RUN AFT THROUGH THE BALANCE AND THE STING BORE TO J0 AT THE ROLL HOUSING.
4. THE MODEL POD (WT-6C-004) IS 3D PRINTED FROM ITS STEP FILE; THE SHEET GIVES THE FIT AND INSPECTION CHECKS.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

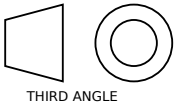
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-001	Balance, 6-component (gauged)	1	7075-T651
2	WT-6C-002	Puck, model interface	1	6061-T6
3	WT-6C-003	Sting	1	4140 28-32 HRC
4	WT-6C-004	Pod, reference model, 2.25 in	1	3D printed (PETG)
5	-	Strain gauge 350 ohm, CEA-13 class	24	Vishay / bought
6	-	Dowel pin 1/8 x 1/2	2	bought
7	-	SHCS #10-32 x 5/8 (puck)	2	bought
8	-	Cross pin 3/16 dowel (sting)	1	bought
9	-	Set screw 1/4-28 cup point	2	bought
10	-	SHCS #8-32 x 3/8 (pod)	4	bought
11	-	Wire 32 AWG PTFE + J0 DC-37	1	see Electronics

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, FLIGHT (BALANCE + STING + MODEL)								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-100		A	B	NTS		-		1 OF 36